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Dynamical Casimir Effect in a Cavity with a Resonantly Oscillating Boundary*

YANG Xiao-Xue[1] and WU Ying[1,2]

[1]Physics Department, Huazhong University of Science and Technology, Wuhan 430074, China

[2]Laboratory of Magnetic Resonance and Atomic and Molecular Physics, Wuhan Institute of Physics and Mathematics, The Chinese Academy of Sciences, Wuhan 430071, China

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Abstract *We present analytical solutions describing quantized vacuum field in a one-dimensional cavity with one of its two mirrors fixed and another vibrating in simple harmonic form. These solutions are accurate up to the second order of the oscillating magnitude and they are uniformly valid for all time. We obtain the simple analytical expression for the energy density of the field which explicitly manifests that for a cavity vibrating at its q-th ($q \geq 2$) eigenfrequency, q traveling wave packets emerge in the finite part of the field energy density, and their amplitudes grow while their widths shrink in time, representing a large concentration of energy. The finite part of the field energy density originating from the oscillations is shown to be proportional to the factor* $(q^2 - 1)$.

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1 Introduction

Over the last several decades, there has been much effort devoted to the quantum theory of the electromagnetic field in a cavity with moving boundaries.[1−12] The topic is of fundamental theoretical interest in that it reveals a number of delicate features of quantum physics such as the dynamical modification of the Casimir force[3] and the vacuum emission of photons with nonclassical photon statistics.[4−7,11] On the other hand, the subject is also of practical importance since it may be closely related to sonoluminescence,[12] high precision optical interferometry,[13] the generation of squeezed light,[14] and quantum nondemolition measurements,[15] etc. Among these investigations, much effort has been focused on the radiation modes of a one-dimensional cavity with one fixed and one oscillating mirrors, particularly when the later oscillates periodically at the eigenfrequencies of the unperturbed cavity modes. Such resonant coupling could greatly distort the vacuum field and lead to pronounced modifications to a number of quantum effects of the vacuum. Theoretical studies of such systems have so far been mainly limited to the small oscillation-amplitude regime by using various small-parameter expansions. For instance, when one of the mirrors oscillates in a simple harmonic form, Dodonov and co-workers have found approximated solution for the case of the second resonance oscillation in both short time[5] and long time[6] regions (They also found out the photon production rates valid throughout the whole time interval from short time up to long time[4]). Dalvit and Mazzitelli[10] have subsequently obtained the approximated solutions for the cases of all q-th resonance oscillations by using a renormalization-group approach which makes the solutions valid for a period of time longer than that of Dodonov *et al.* However, all of these solutions are valid only up to the first order of the ratio of the maximum speed of the oscillating mirror to the speed of light in vacuum. Cole and Schieve[8] have recently developed a simple and elegant method to calculate numerically the dynamics of the quantum vacuum field in a one-dimensional cavity with an arbitrarily moving boundary. On the other hand, Law[7] has obtained the first exact solution for a nearly simple harmonic oscillating mirror with the second unperturbed eigenfrequency. We[16] have recently obtained the exact analytical solutions for a nearly simple harmonic oscillating mirror with any one of the unperturbed eigenfrequencies, and have claimed that the solutions presented in that paper reflect all the main generic features for all possible kinds of resonant vibrations of a one-dimensional cavity. However, the trajectories of the moving mirror considered in that paper[16] are somewhat strange and it is hence interesting to consider directly the situations where one of the mirrors oscillates in a simple harmonic form.

In this article, we present a series of approximated analytical solutions accurate up to the second order of the ratio of the maximum speed of the oscillating mirror to the speed of light in vacuum, and they are uniformly valid for all time when one of the mirrors oscillates in a simple harmonic form, with each solution corresponding to a particular unperturbed eigenfrequency at which the mirror oscillates. Our results in this paper also confirm the conclusion that the solutions presented in Ref. [16] indeed reflect all the main generic features for all possible kinds of resonant vibrations of a one-dimensional cavity.

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In addition they are compared in details with the previous first-order results on the same problem.[5,6,10]

2 Model and Solutions

We consider the dynamics of the quantum vacuum in a one-dimensional cavity formed by two ideal mirrors: one fixed at $x = 0$, the other moving in a prescribed trajectory $x = L(t)$, where $L(t \leq 0)$ is assumed to be a constant L_0. The field quantization of this system is done by expressing the vector potential as[1]

$$A(x,t) = \sum_{k=1}^{\infty} [a_k \phi_k(x;t) + a_k^\dagger \phi_k^*(x;t)] \,, \tag{1}$$

where a_k and $a_k^\dagger$ are the time-independent annihilation and creation operators defined for the mode structure existing for $t \leq 0$. The mode functions $\phi_k(x;t)$ are given in terms of an auxiliary function $R(y)$ as follows:

$$\phi_k(x;t) = \frac{\mathrm{i}}{\sqrt{4\pi k}} (\, \mathrm{e}^{-\mathrm{i}k\pi R(y_+)} - \mathrm{e}^{-\mathrm{i}k\pi R(y_-)}) \,, \tag{2}$$

where $y_\pm \equiv t \pm x$ and we have taken $c = \hbar = 1$ for convenience. Here $R(y)$ is a real function which satisfies

$$R(t + L(t)) = R(t - L(t)) + 2 \,, \tag{3}$$

and $R\,(y \leq L_0) = y/L_0$. A unique solution for $R(y > 0)$ is then defined by the recursion relation (3) once a mirror trajectory $L(t \geq 0)$ is given. Here we consider a family of mirror trajectories described by

$$L(t \geq 0) = L_0 \Big[1 + \epsilon \sin\Big(\frac{q\pi t}{L_0} \Big) \Big] \,, \quad q = 1, 2, 3, \cdots \,, \tag{4}$$

where we assume $|\epsilon_q| \equiv |(-1)^q q\pi\epsilon| \ll c = 1$, i.e., the maximum speed of the moving mirror is assumed to be much less than the speed of light in vacuum.

In order to obtain the solution of Eq. (3), let $t + L(t) = 2nL_0 + z_+$ where n is a non-negative integer so that $z_+ \in (-L_0, L_0]$, one then arrives at $t - L(t) = 2(n-1)L_0 + z_-$ and

$$\frac{q\pi z_\pm}{L_0} = \Delta_q \pm \epsilon_q \sin \Delta_q \,, \tag{5}$$

where $\Delta_q = (q\pi/L_0)[t - (2n-1)L_0]$ and $\epsilon_q = (-1)^q q\pi\epsilon$. Utilizing the identity (see appendix A)

$$\begin{aligned} \frac{\tan(\alpha - \beta)}{\tan(\alpha + \beta)} &= \frac{1 - (2\tan\beta/\sin 2\alpha) + \tan^2(\beta)}{1 + (2\tan\beta/\sin 2\alpha) + \tan^2(\beta)} \\ &= \frac{1 - (2\beta/\sin 2\alpha) + \beta^2}{1 + (2\beta/\sin 2\alpha) + \beta^2} + O(\beta^3) \,, \end{aligned} \tag{6}$$

and taking $2\alpha = \Delta_q$, $2\beta = \epsilon_q \sin\Delta_q$, one yields

$$\frac{\tan(q\pi z_-/2L_0)}{\tan(q\pi z_+/2L_0)} = \frac{1 - \epsilon_q}{1 + \epsilon_q} + O(\epsilon_q^3) \,,$$

where use has been made of

$$\frac{1 - \epsilon_q + \beta^2}{1 + \epsilon_q + \beta^2} = \frac{1 - \epsilon_q}{1 + \epsilon_q} + \frac{2\epsilon_q \beta^2}{(1 + \epsilon_q)^2} + \cdots \,.$$

Note that all the neglected terms in the above equations are of the form $O(\epsilon_q^k \sin^l(q\pi t/L_0)), l = 2, 3, \cdots; k = 3, 4, \cdots$ (Note $\sin\Delta_q \equiv (-1)^q \sin(q\pi t/L_0)$). It is then straightforward to see that the approximated solutions have the form (See also appendix A)

$$\begin{aligned} R_q(2nL_0 + z) = 2n + \frac{2}{q\pi} \tan^{-1} \Big[\Big(\frac{1 - \epsilon_q}{1 + \epsilon_q} \Big)^n \tan\Big(\frac{q\pi z}{2L_0} \Big) \Big] \\ + O(\epsilon_q^3) \,, \end{aligned} \tag{7}$$

where the branch of the multivalue function $\tan^{-1} x$ should be carefully chosen to keep R continuous, $\epsilon_q = (-1)^q q\pi\epsilon$, $z \in (-L_0, L_0]$, $n = 0, 1, 2, \cdots$, and $q = 1, 2, 3, \cdots$. We have checked our analytic solutions in Eq. (7) for some of the trajectories in Eq. (4) with numerical solutions using the method in Ref. [8] and found excellent agreement. Here we do not present the figures describing the function R versus z in Eq. (7) since they are similar to those in Fig. 1 of Ref. [16]. Our solutions obviously satisfy the boundary condition $R(z) = z/L_0$, $z \in (-L_0, L_0]$. Compared with the solutions obtained by Dalvit *et al.* by using the renormalization-group method,[10] our solutions are much simpler than theirs, which makes crucial difference in obtaining the analytical expression of the energy density of the quantum vacuum field as will be seen shortly. What is more, our solutions are accurate up to $O(\epsilon_q^2)$ while theirs and those of Dodonov *et al.*[5,6] are accurate only up to $O(\epsilon_q)$ where $|\epsilon_q|$ represents the ratio of the maximum speed of the moving mirror to the speed of light in vacuum. The last essential difference originates from the fact that all the neglected terms in our solutions are of the form $O(\epsilon_q^k \sin^l(q\pi t/L_0))$, $l = 2, 3, \cdots$, $k = 3, 4, \cdots$; while they[10] neglected the terms of the order $O(\epsilon_q^2 t/L_0)$. Therefore, their solutions are valid only for time t satisfying $\epsilon_q^2 t/L_0 \ll 1$, while ours are uniformly valid for all time so long as $\epsilon_q^3 \ll 1$. In the appendix B, we present the more detailed comparison of our solutions with the previous ones.

3 Vacuum Energy Density

Once the solution $R(z)$ is known, the vacuum energy density can be calculated by using the general formula derived by Fulling and Davies.[9] The cutoff-independent expression of the vacuum energy is[7,8]

$$\langle T_{00}(x,t) \rangle_{\mathrm{reg}} = -f(t+x) - f(t-x) \,, \tag{8}$$

$$24\pi f = \frac{R'''}{R'} - \frac{3}{2} \Big(\frac{R''}{R'} \Big)^2 + \frac{\pi^2}{2} (R')^2 \,, \tag{9}$$

where R', R'', etc., are the derivatives of R. Substituting the form R_q into this equation, we obtain the corresponding f (denoted f_q) as follows (The derivation is given in appendix B):

$$f_q(2nL_0 + z) = -\frac{q^2}{2} E_{\mathrm{sct}} - \frac{(q^2 - 1)\pi}{48 D_{n;q}^2(z) L_0^2} \,, \tag{10}$$

or its equivalent form

$$f_q\,(2nL_0+z) = -\frac{1}{2}E_{\rm sct} - \frac{(q^2-1)\pi}{48L_0^2}\Big[\frac{1}{D^2_{n;q}(z)} - 1\Big], \quad (11)$$

where $E_{\rm sct} = -\pi/24L_0^2$ is the static Casimir term which is present even when the mirrors are at rest, and

$$D_{n;q}(z) = \frac{1}{2}\Big[(G^n + G^{-n}) - (G^n - G^{-n})\cos\Big(\frac{q\pi z}{L_0}\Big)\Big] \quad (12)$$

with $G = (1-\epsilon_q)/(1+\epsilon_q)$. Note that the final expression (10) of the energy density for our approximated solutions corresponding to the mirror oscillating in the simple harmonic case is very simple and has the same form as the one for the exact solutions obtained by us[16] corresponding to the mirror oscillating in a nearly simple harmonic case except for different $D_{n,q}(z)$. While it is, if not impossible, very difficult to obtain the analytical expression of the energy density in a simple form by using the solutions of Dalvit *et al.* In Figs 1 and 2, we have plotted some snapshots of the dimensionless regular energy density $T \equiv \langle T_{00}(x,t)\rangle_{\rm reg}/|E_{\rm sct}|$ for the harmonic orders $q = 3$ and $q = 4$ respectively.

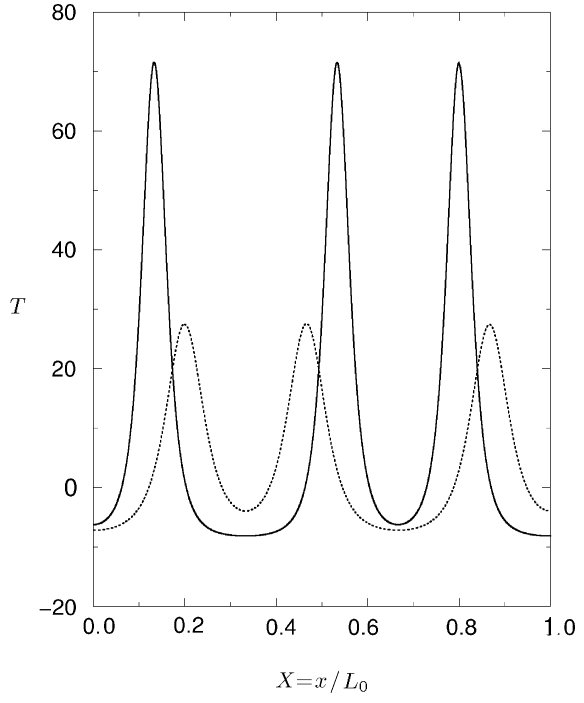


Fig. 1 Snapshots of the dimensionless regular energy density $T \equiv \langle T_{00}(x,t)\rangle_{\rm reg}/|E_{\rm sct}|$ versus $X = x/L_0$ for the harmonic order $q = 3$ and the maximum speed $|\epsilon_3| = 0.001c$ at time instances $ct/L_0 = 1100.2$ (dotted line) and $ct/L_0 = 1500.8$ (solid line).

Let us discuss the energy density expressed in Eq. (10) or Eq. (11). The first term on the right-hand side of Eq. (10) is independent of the space-time variable and determines the long-time energy density in the so-called sub-Casimir region, i.e., the region far from those of the wave packets.[7] It contributes $q^2E_{\rm sct}$ to the vacuum energy density. The second term on the right-hand side of Eq. (10) for $q \geq 2$ describes a coherent structure exhibited as narrow wave packets in the energy density. There exists no wave packet at all for the special case of $q = 1$, which can be most easily seen from Eq. (11). It is pointed out that all the effects of the oscillating boundary on the energy density of the field modes are proportional to the factor (q^2-1), and are contained in the second term on the right-hand side of Eq. (11). This fact clearly manifests the reluctance of the vacuum to respond to the fundamental resonant perturbation ($q = 1$). In other words, the fundamental resonant perturbation is not effective in creating photons and in enhancing the Casimir force. Let us now discuss the features of the traveling wave packets existed as $q \geq 2$. A straightforward analysis of the function $D_{n;q}(z)$ shows that for a given value of n and $q \geq 2$, f_q has q identical peaks located within the interval $z \in (-L_0, L_0)$, representing q wave packets traveling back and forth inside the cavity with the speed of light. It is easily shown from Eq.(10) that the amplitudes of the q traveling wave packets grow while their widths shrink in time. The height H of all the q peaks is easily seen to be

$$H = q^2E_{\rm sct} + \frac{(q^2-1)\pi}{48L_0^2}\Big(\frac{1+|\epsilon_q|}{1-|\epsilon_q|}\Big)^{2n}, \quad (13)$$

where we have neglected the influences of traveling wave packets on each other, i.e., choosing any particular wave packet, say, the one located at position x_1 at certain instant time, the value of energy density at the same position caused by other wave packet(s) is not included in Eq. (13). This equation explicitly displays that the amplitudes of the traveling wave packets grow as time (characterized by the integer n) increases. The width W of all the q wave packets is easily shown to shrink in time according to the form $W/L_0 \propto [(1-|\epsilon_q|)/(1+|\epsilon_q|)]^n/q$ for sufficiently large n, implying that we could introduce the concept of the localized photons in long-time asymptotic regime. As a matter of fact, the peak height of the wave packets $H \approx [(q^2-1)\pi/(48L_0^2)][(1+|\epsilon_q|)/(1-|\epsilon_q|)]^{2n}$ for sufficiently large n can be explained based on the Doppler effect of photons traveling back and forth within the oscillating cavity.[17] Before ending this paragraph, we would like to mention that due to the staircase structure of the function R_q in Eq. (7), the numerical computations on the energy density of the oscillating cavity may introduce some fictitious outcomes on the wave packets which relate to the jumps of the function R_q. For instance, there exists no wave packet for the special case of $q = 1$ at least up to the order $|\epsilon_1|^2$ even though $D^{-2}_{n;q=1}(z)$ does have a maximum value $[(1+|\epsilon_1|)/(1-|\epsilon_1|)]^{2n}$ within the interval $z \in (-L_0, L_0)$. From the expression of function R_q in Eq. (7), it is easily seen that a staircase structure

does develop in $R_1(y)$ just as that in other R_q for $q \geq 2$, and as $y \to \infty$, it approaches a "perfect" staircase with a height of 2, the highest jump height of all R_q (Note that the jump height is $2/q$). However, the combination of the derivatives R_1', R_1'' and R_1''' in the expression of the vacuum energy density happens to cancel out the effect of the jump in R_1, resulting in no traveling wave packet at all. However, a wave packet for $q = 1$ may emerge if small numerical errors are introduced. This can be understood by noting that the cancellation of the jump effect for $q = 1$ ceases to be exact if some or all of the derivatives R_1', R_1'' and R_1''' are slightly modified. By the way, we point out that figure 2 in Ref. [10] for energy density profile has error according to our results. It is noted that all q traveling wave packets within the cavity for $q \geq 2$ should have nearly identical height accurate up to the order ϵ_q^2 if they are either well apart from each other or equally spaced according to Eq. (13) which is calculated based on our solutions (7) accurate up to the same order ϵ_q^2. However, in that figure,[10] one (the rightmost one) of the four traveling waves well apart from each other has the height over value 230 and the heights of the rest three are all under value 150, too large a difference to be accounted for by either $O(\epsilon_q^2)$ (Their solutions are valid up to $O(\epsilon_q)$) correction or the unavoidable numerical errors. In this regard, our analytical expression of the energy density may also be helpful to numerical computations on the oscillating cavity.

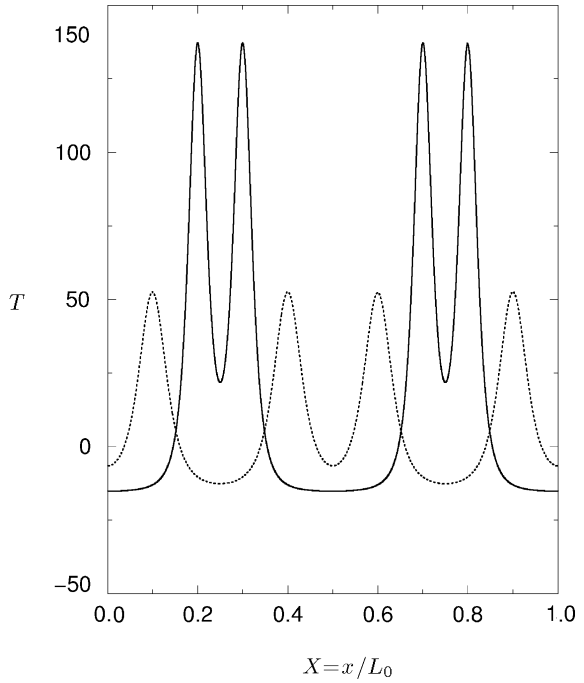


Fig. 2 Snapshots of the dimensionless regular energy density $T \equiv \langle T_{00}(x,t)\rangle_{\rm reg}/|E_{\rm sct}|$ versus $X = x/L_0$ for the harmonic order $q = 4$ and the maximum speed $|\epsilon_4| = 0.001c$ at time instances $ct/L_0 = 1100.1$ (dotted line) and $ct/L_0 = 1500.8$ (solid line).

4 Conclusion

In summary, we have presented a family of analytical solutions for the quantum vacuum field in a one-dimensional cavity with one of the mirrors oscillating in a simple harmonic form with any one of its unperturbed eigenfrequencies. These analytical solutions are accurate up to the second order of the ratio of the oscillating speed to the speed of light in vacuum and they are uniformly valid for all time. We have also thoroughly compared ours with the previous solutions to the same problem. Our solutions not only have the higher accuracy (up to $O(\epsilon_q^2)$) than the accuracy of the previous known solutions (accurate only up to $O(\epsilon_q)$) but also are uniformly valid for all time while the previous known solutions suffer from the limitation on time t ($t\epsilon_q^2 \ll 1$). In addition, an important advantage of our solutions over the previous ones is the simplicity in form. This merit permits us to obtain simple analytical expression for energy density of the quantized vacuum field, which clearly manifests the intriguing features of energy density within a cavity in responding to resonant perturbations. Our results display explicitly that for a cavity with one of the mirrors vibrating in a simple harmonic form at its q-th ($q \geq 2$) unperturbed eigenfrequency, q traveling wave packets emerge in the finite part of the field energy density with their amplitudes growing while their widths shrinking in time, and there exists a "sub-Casimir" region where the energy density equals q^2 times the static Casimir energy density. The field energy density due to the oscillating boundary is shown to be proportional to the factor $(q^2 - 1)$. The solutions in this paper can be utilized to calculate the photon production rates, and the corresponding squeezing and correlation coefficients for a cavity with a harmonic oscillating mirror up to $O(\epsilon_q^2)$, which will be discussed elsewhere.

Appendix A

In this appendix, we prove Eqs (6) and (11), and explain that for $B = O(\epsilon^3)$, $\arctan(A + B) = \arctan A + O(\epsilon^3)$, one of the properties utilized to obtain Eq. (7).

We begin with the proof of $\arctan(A+B) = \arctan A + O(\epsilon^3)$ as $B = O(\epsilon^3)$. Let $F(x) = \arctan x$, and utilize $F(A + B) = F(A) + F'(A)B + F''(A)B^2 + \cdots$ (Taylor expansion around point A), and note that $F'(A) = 1/(1 + A^2)$ and all the other derivatives F'', F''', etc. are finite quantities, one is then straightforward to see that $\arctan(A + B) = \arctan A + O(\epsilon^3)$ as $B = O(\epsilon^3)$.

Next, we prove Eq. (6). Using $2\sin A\cos B = \sin(A + B) + \sin(A - B)$ and $\sin 2\beta = 2\tan\beta/(1 + \tan^2\beta)$, we obtain

$$\frac{\tan(\alpha-\beta)}{\tan(\alpha+\beta)} = \frac{\sin(\alpha-\beta)\cos(\alpha+\beta)}{\cos(\alpha-\beta)\sin(\alpha+\beta)} = \frac{\sin(2\alpha)-\sin(2\beta)}{\sin(2\alpha)+\sin(2\beta)}$$

$$= \frac{\sin(2\alpha) - 2\tan\beta/(1+\tan^2\beta)}{\sin(2\alpha) + 2\tan\beta/(1+\tan^2\beta)}$$
$$= \frac{1 - (2\tan\beta/\sin 2\alpha) + \tan^2(\beta)}{1 + (2\tan\beta/\sin 2\alpha) + \tan^2(\beta)}\,, \qquad \text{(A1)}$$

which results in the first equality sign in Eq. (6). Taking $2\alpha = \Delta_q$ and $2\beta = \epsilon_q \sin\Delta_q$, and noting $\tan\beta = \beta + O(\beta^3) = \beta + O(\epsilon_q^3)$, one yields

$$\frac{\tan(\alpha-\beta)}{\tan(\alpha+\beta)} = \frac{1 - (2\beta/\sin 2\alpha) + \beta^2}{1 + (2\beta/\sin 2\alpha) + \beta^2} + O(\beta^3)$$
$$= \frac{1 - \epsilon_q + \beta^2}{1 + \epsilon_q + \beta^2} + O(\beta^3)$$
$$= \frac{1-\epsilon_q}{1+\epsilon_q} + \frac{2\epsilon_q\beta^2}{(1+\epsilon_q)^2} + O(\epsilon_q^3)\,, \qquad \text{(A2)}$$

where we have made use of the expansion $(a+x)/(b+x) = a/b + (b-a)x/b^2 + \cdots$ with $a = 1-\epsilon_q$, $b = 1+\epsilon_q$, and $x = \beta^2 = O(\epsilon_q^2)$. Equation (A2) gives the derivations of the second equality sign in Eq. (6), and the two equations between Eqs (6) and (7).

We now present the derivation of Eq. (11) for the expression of the energy density. Let $g = R'$ and note $(g'/g)' = (gg'' - g'^2)/g^2 = g''/g - (g'/g)^2$, equation (9) can be put into the form

$$24\pi f = \Big(\frac{g'}{g}\Big)' - \frac{1}{2}\Big(\frac{g'}{g}\Big)^2 + \frac{\pi^2}{2}g^2\,. \qquad \text{(A3)}$$

Equation (7), after neglecting $O(\epsilon_q^3)$ terms, is rewritten here as

$$R(2nL_0 + z) = 2n + \frac{2}{q\pi}\tan^{-1}\Big[G^n \tan\Big(\frac{\psi}{2}\Big)\Big]\,, \qquad \text{(A4)}$$

where $R \equiv R_q$, $G = (1-\epsilon_q)/(1+\epsilon_q)$, and $\psi \equiv \psi(z) = (q\pi z)/L_0$. For any fixed integer n, the superscript "$\prime$" means the derivative with respect to the variable z, and in particular,

$$g = \frac{\mathrm{d}R}{\mathrm{d}z} = \frac{2}{q\pi}\frac{G^n}{1 + G^{2n}\tan^2(\psi/2)}\frac{q\pi}{2L_0\cos^2(\psi/2)}$$
$$\equiv \frac{1}{L_0 D}\,, \qquad \text{(A5)}$$

where D is denoted by $D_{n;q}(z)$ in the main text and has the form

$$D = G^{-n}\cos^2(\psi/2) + G^n \sin^2(\psi/2)$$
$$= \frac{1}{2}[(G^n + G^{-n}) - (G^n - G^{-n})\cos\psi]$$
$$\equiv \cosh k - \sinh k\cos\psi\,, \qquad \text{(A6)}$$

where $\cosh k = (G^n + G^{-n})/2$, $\sinh k = (G^n - G^{-n})/2$, and use has been made of $\cos^2(\psi/2) = (1+\cos\psi)/2$ and $\sin^2(\psi/2) = (1-\cos\psi)/2$. Then one sees

$$\frac{g'}{g} = -\frac{D'}{D} = -\frac{q\pi}{L_0}\frac{\sinh k\sin\psi}{D}\,,$$
$$\Big(\frac{g'}{g}\Big)' = -\Big(\frac{q\pi}{L_0}\Big)^2 \frac{D\sinh k\cos\psi - \sinh^2 k\sin^2\psi}{D^2}\,,$$

and hence

$$\Big(\frac{g'}{g}\Big)' - \frac{1}{2}\Big(\frac{g'}{g}\Big)^2 = -\frac{q^2\pi^2 W}{2L_0^2 D^2}\,,$$

where

$$W \equiv [2D\sinh k\cos\psi - 2\sinh^2 k\sin^2\psi + \sinh^2 k\sin^2\psi]$$
$$= 2\sinh k\cosh k\cos\psi - 2\sinh^2 k(\sin^2\psi + \cos^2\psi) + \sinh^2 k\sin^2\psi$$
$$= 2\sinh k\cosh k\cos\psi - 2\sinh^2 k + \sinh^2 k(1-\cos^2\psi)$$
$$= 1 - [\cosh^2 k - 2\sinh k\cosh k\cos\psi + \sinh^2 k\cos^2\psi] \equiv (1 - D^2)\,, \qquad \text{(A7)}$$

where the identity $\sinh^2 k = \cosh^2 k - 1$ has been used. Therefore, we obtain

$$24\pi f = \frac{\pi^2}{2L_0^2} - \frac{(q^2-1)\pi^2}{2L_0^2}[D^{-2} - 1]\,, \qquad \text{(A8)}$$

which is Eq. (11) except for different notation.

Appendix B

In this appendix, we present a detailed comparison of our solutions with the previous ones. In order to further compare our solutions up to the $O(\epsilon_q^2)$ accuracy with the previous ones accurate only up to the $O(\epsilon_q)$, we use our solutions (7) to derive several forms accurate only up to $O(\epsilon_q)$. Noting that $[(1-\epsilon_q)/(1+\epsilon_q)]^n = \exp(-2n\epsilon_q) + O(n\epsilon_q^2)$,

$$\tan\Big[\frac{q\pi}{2}\Big(\frac{z}{L_0} - \frac{2}{q\pi}\tan^{-1}\Psi\Big)\Big] = \exp(-2n\epsilon_q)\tan\Big(\frac{q\pi z}{2L_0}\Big)\,,$$

where $z = y - 2nL_0$ and

$$\Psi \equiv \frac{\sin(q\pi y/L_0)}{\coth(n\epsilon_q) + \cos(q\pi y/L_0)}\,,$$

one easily arrives at from Eq. (7),

$$R_q(y = 2nL_0 + z) = \frac{y}{L_0} - \frac{2}{q\pi}\tan^{-1}\Big[\frac{\sin(q\pi y/L_0)}{\coth(n\epsilon_q) + \cos(q\pi y/L_0)}\Big] + O(n\epsilon_q^2)\,, \qquad \text{(A9)}$$

which can be put into another equivalent form

$$R_q(y = 2nL_0 + z) = \frac{y}{L_0} - \frac{2}{q\pi}\mathrm{Im}\ln\Big[1 + \xi + (1-\xi)\exp\Big(\frac{q\pi y}{L_0}\Big)\Big] + O(n\epsilon_q^2)\,, \qquad \text{(A10)}$$

where $\xi = \exp(-2n\epsilon_q)$ and $\epsilon_q = (-1)^q q\pi\epsilon$. Note that these two forms are valid for all time bounded above by the condition $n\epsilon_q^2 \ll 1$ or $(t/L_0)\epsilon_q^2 \ll 1$ by noting that time t relates n by $t = (2n-1)L_0 + \eta$ and $\eta \in (-L_0, L_0]$ (See the definition of Δ_q after Eq. (5)). It is interesting to note that once $2n$ in ξ is replaced by $y/L_0 \equiv (2n + z/L_0)$ where $z \in (-L_0, L_0]$, equation (9) for $q = 2$ is the same as the long-time asymptotical solution obtained by Dodonov and Klimov[6] (See Eq. (21) in Ref. [6]). This fact carries implications. First of all, we have here proved that such a replacement makes their long-time asymptotical solution throughout the time interval $[0, t_m]$ where t_m satisfying $(t_m/L_0)\epsilon_q^2 \ll 1$. Note that although they[6] have already found that the brute short-time expansion of their long-time asymptotical solution (obtained under the assumption of $\epsilon \ll 1$ and $\epsilon t \gg 1$) happens to coincide with their short-time approximated solution,[5] no one has known so far whether or not the solution is valid between these two time limits. Secondly, such a replacement is critical to make their solution satisfy the correct initial condition $L(t \le 0) = L_0$ or equivalently the correct boundary condition $R(z) = z/L_0,\ z \in (-L_0, L_0]$. To see this, let us consider the short-time limit where the exact meaning of the short time is the integer n satisfying $n|\epsilon_q| \ll 1$. In this case, it is easy to obtain from Eq. (A9) or Eq. (A10) that $R(y = 2nL_0 + z) \approx y/L_0 - (-1)^q 2n\epsilon \sin(q\pi y/L_0)$ which satisfies the correct initial condition and is identical to the solution obtained by Dalvit *et al.* in the short-time limit. If $2n$ in this expression is replaced by y/L_0, it results in the short-time solution presented in Ref. [5] (equations (11) and (15) there) which does not satisfy the correct initial condition as pointed out by Dalvit *et al.*[10] Lastly, it is seen from the above discussion that the solution accurate up to the order ϵ_q obtained by Dodonov and Klimov[6] is in fact valid throughout the time interval $[0, t_m]$ only by a very small modification and hence, due to its simplicity, is superior to the cumbersome expression of the solutions in Ref. [10] which are valid in the same time interval and up to the same order, i.e., the first order of ϵ_q.

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